

LEVELLING:-

- The aim of levelling is to determine the relative heights of different objects on or below the surface of earth and to determine the undulation of the ground surface.

levelling is done for the following purposes:

- a) To prepare a contour map for fixing sites for reservoirs, dams etc and fix the alignment of roads, railways, irrigation canals etc.
- b) To determine the altitudes of different important points on a hill or to know the reduced levels of different points on or below the surface of the earth.
- c) To prepare a longitudinal section and cross-sections of a project in order to determine the volume of earth work.
- d) To prepare a layout map for water supply, sanitary or drainage schemes.

LEVELLING:-

- Art of determining the relative heights of different points. Levelling deals with measurement in the vertical plane.

LEVEL SURFACE:

- Any surface parallel to the mean spheroidal surface of the earth is said to be level surface. Such a surface is obviously curved. The water surface of a still lake is also considered a level surface.

LEVEL LINE:-

Any line lying on a level surface is called level line. This line is normal to the plumb line (direction of gravity) at all points.

HORIZONTAL PLANE:

Any plane tangential to the level surface at any point is known as the horizontal plane.

HORIZONTAL LINE:

Any line lying on the horizontal plane is said to be horizontal line. It is a straight line tangential to the level line.

VERTICAL PLANE:

Any plane passing through the vertical line is known as the vertical plane.

VERTICAL LINE:

The direction indicated by the plumb line (the direction of gravity) is known as the vertical line.

DATUM SURFACE OR LINE:

This is an imaginary level surface or level line from which the vertical distances (above or below this line) of different points are measured.

In ~~the~~ India, datum adopted for the Great Trigonometrical Survey (GTS) is the Mean Sea Level (MSL) at Mumbai.

REDUCED LEVEL (R.L.):

The vertical distance of a point above or below the datum line is known as the reduced level. The RL of a point may be positive or negative accordingly as the point is above or below the datum.

LINE OF COLLIMATION:

It is an imaginary line passing through the intersection of the cross-hairs at the diaphragm and the optical centre of the object glass and its continuation. Line of collimation is also known as line of sight.

BENCH MARKS:-

- These are fixed points of known elevation.
- Bench marks serve as reference points for finding the RL of new points or for conducting levelling operations in projects involving roads, railways etc.
- Bench marks may be of four types:
 - a) GTS bench marks [Great Trigonometric Survey]
 - b) Permanent bench marks
 - c) Temporary bench marks
 - d) Arbitrary bench marks.

GTS Bench marks:

- Established by the Survey of India Department at large intervals all over the country.
- The values of reduced level, the relevant positions and the number of bench marks are given in a catalogue published by this department.



Marking on brass plate.

PERMANENT BENCH MARKS:

- These are fixed points or marks established by different Government departments like PWD, Railways, Irrigation etc.
- The RLs of these points are determined with reference to the GTS bench-mark and are kept on permanent points like the plinth of a building, parapet or culvert and so on. [Some times underground pillars]

ARBITRARY BENCH MARKS:

- When the RLs of some fixed points are assumed, they are termed arbitrary bench-marks.
- These are adopted in small survey operations when only the undulation of the ground surface is required to be determined.

TEMPORARY BENCHMARKS:

- Benchmarks established temporarily at the end of day's work.
- Root of a tree, parapet of a nearby culvert, a furrow post.

LEVELLING OPERATIONS:

Back sight reading (BS):

- First reading taken after setting up of the instrument (Level)
- This reading is always taken on a point of known RL (Bench mark or change point)
- The readings are taken on a graduated levelling staff.

FORE SIGHT READING:- (FS)

It is the last staff reading in any set-up of the instrument, and indicates the shifting of the later.

INTERMEDIATE SIGHT READING: (IS)

It is any other staff reading between the BS and FS in the same set up of the instrument.

CHANGE POINT: (CP):

This point indicates the shifting of the instrument. At this point, FS is taken from the current instrument position and BS is taken from the next instrument setting.

TYPES OF LEVEL:

Dumpy Level: - Telescope is rigidly fixed to its supports.

- Instrument is stable
- Commonly used
- Can't be rotated about its longitudinal axis

WYE LEVEL: - Telescope is held in two 'Y' supports.

- Can be removed from the supports.
- Telescope can be rotated.

Cooke's Reversible }
Level } : - Combination of dumpy level & Y level

- Supported by two rigid sockets
- Telescope can be rotated
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CUSHING'S LEVEL: - Telescope cannot be removed from sockets

- The eye piece and object glass are removable & interchangeable

MODERN TILTING }
LEVEL } : - Telescope can be tilted about its horizontal axis

- Tilting screw helps in making line of collimation horizontal.

Automatic Level:

- Self-aligning level
- Instrument is automatically levelled within a certain tilt range by means of a compensating device (tilt compensator).

TYPES OF LEVELLING STAFF:

Target staff: - Movable target.

- Target is provided with a vernier which is adjusted by the Staffman, as per directions of level man.
- Used for long sightings.

Self-reading staff:

Telescopic Staff: Top portion - Solid - 1.25 m length
 Central box - hollow - 1.25 m length
 Bottom - hollow - 1.5 m length

Folding Metric staff: Well-seasoned timber ; 75 mm width ;
 18 mm Thickness ; 4 m length.

One length staff: 3m long.

Invar staff: - Precise levelling - 3 m long.

PROBLEMS:-

Staff Position	Chainage (m)	B.S (m)	I.S (m)	F.S (m)	H.I (m)	Rise (m)	Fall (m)	R.L (m)	Remarks
1	165	3.150			101.235			98.085	B.M
2	180		2.245			0.905		98.990	
3	195		1.125			1.120		100.110	
4	210	3.125		0.860	103.500	0.265		100.375	C.P
5	225		2.760			0.365		100.740	
6	240		1.835			0.925		101.665	
7	255		1.470			0.365		102.030	
8	270	1.225		1.965	102.760		0.495	101.535	C.P
9	285		2.390				1.165	100.370	
10	300			3.035			0.645	99.725	

Arithmetic check:

$$\sum BS - \sum FS = 7.500 - 5.860 = +1.640$$

$$\text{Last RL} - \text{1st RL} = 99.725 - 99.085 = +1.640$$

$$\sum \text{Rise} - \sum \text{Fall} = 3.945 - 2.305 = +1.640$$

- 2). The following successive readings were taken with a dumpy level along a chain line at common intervals of 20 m. The first reading was taken on a chainage of 140 m. The RL of the second ~~change~~ change point was 107.215 m. The instrument was shifted after the third & seventh readings. Calculate the RLs of all the points. [3.150, 2.245, 1.125, 3.860, 2.125, 0.760, 2.235, 0.470, 1.935, 3.225, 3.890 m]

Staff Position	Chainage (m)	BS (m)	IS (m)	FS (m)	Rise (+) (m)	Fall (-) (m)	RL (m)	Remarks
1	140	3.150					103.565	
2	160		2.245		0.905		104.470	
3	180	3.860		1.125	1.120		105.590	1 st C.P
4	200		2.125		1.735		107.325	
5	220		0.760		1.365		108.690	
6	240	0.470		2.235		1.475	107.215	2 nd C.P
7	260		1.935			1.465	105.750	
8	280		3.225			1.290	104.460	
9	300			3.890		0.665	103.795	

CHECK:

$$\sum BS - \sum FS = \text{Last RL} - \text{First RL} = \sum \text{Rise} - \sum \text{Fall}$$

3) The following consecutive readings were taken with a level and a 4 metre levelling staff on a continuously sloping ground at common intervals of 20 m.

0.855(A), 1.545, 2.335, 3.115, 3.825, 0.455, 1.380, 2.055, 2.855, 3.455, 0.585, 1.015, 1.850, 2.755, 3.845 (on B)

The RL of A is 380.500. Make entries in a level book and apply the usual checks. Determine the gradient of AB.

Staff Posi.	BS	IS	FS	Rise	Fall	R.L	Remarks	chainage.
A	0.855					380.500		0
A1		1.545			0.690	379.810		30
A2		2.335			0.790	379.020		60
A3		3.115			0.780	378.240		120
A4	0.455	2.335	3.825		0.710	377.530	C.P	150
A5		1.380			0.925	376.605		180
A6		2.055			0.675	375.930		210
A7		2.855			0.800	375.130		240
A8	0.585		3.455		0.600	374.530	C.P	270
A9		1.015			0.430	374.100		300
A10		1.850			0.835	373.265		330
A11		2.755			0.905	372.360		360
B			3.845		1.090	371.270		

$$\sum BS - \sum FS = \sum Rise - \sum Fall = \text{Last R.L} - \text{First R.L} = -9.230$$

$$\text{Falling gradient of AB} = \frac{\text{difference in level}}{\text{Horizontal distance}} = \frac{9.230}{360} = \frac{1}{39}$$

1 in 39

4) In fly levelling from a BM of RL 140.605, the following readings were observed.

Backsight - 1.545, 2.695, 1.415, 2.925

Foresight - 0.575, 1.235, 0.595

From the last position of the instrument, six pegs at 20 m intervals are to be set out on a uniformly rising gradient of 1 in 50, the first peg is to have an RL of 144.000. Find the staff readings and RLs of the pegs.

Staff-Position	Chainage	BS	IS	FS	Rise (+)	Fall (-)	RL	Remarks
1		1.545					140.605	B.M.
2		2.695		0.575	0.970		141.575	
3		1.415		1.235	1.460		143.035	
4		2.925		0.595	0.820		143.855	
Peg 1	0		2.780		0.145		144.000	B.M. supplied
Peg 2	20		2.380		0.400		144.400	
Peg 3	40		1.980		0.400		144.800	
Peg 4	60		1.580		0.400		145.200	
Peg 5	80		1.180		0.400		145.600	
Peg 6	100			0.780	0.400		146.000	

$$\text{Rise per 20 m} = \frac{20}{50} = 0.400$$

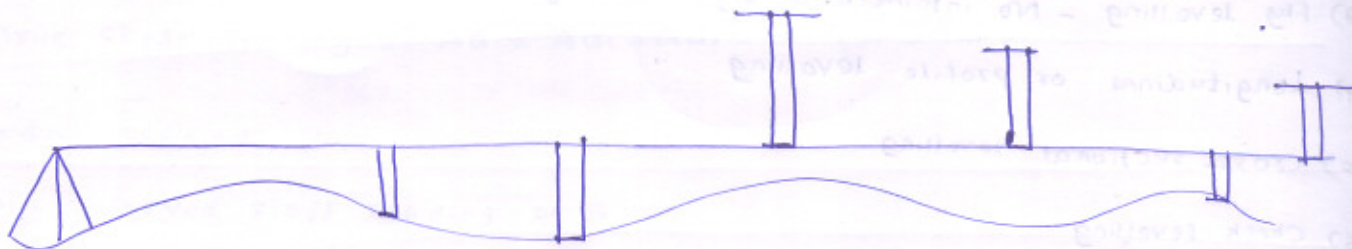
$$50 \rightarrow 1$$

$$20 \rightarrow$$

$$\sum B.S. - \sum F.S. = \sum \text{rise} - \sum \text{Fall} = \text{last RL} - \text{First RL}$$

5) The following readings were taken with a level of auto level and 4-m folding levelling staff- [~~by~~ ~~level~~]

1.270, 1.330, -2.000, -1.500, 1.200, -1.300



Staff position	B.S	F.S	I.S	Rise	Fall	R.L	Remarks
1	1.270						
2		1.330				100.000	B.M.
3	-1.500		-2.000	2.330	0.060	99.94	
4		1.200			2.700	103.27	C.P.
5			-1.300	2.500	2.	100.57	
						103.07	C.P.



α = Angle of inclination of collimation line

A_a = true reading.

A_{a1} = Observed staff reading on A.

$$\text{Error} = A_{a1} - A_a = a_{a1} = D \tan \alpha$$

$$\therefore \text{True staff reading} = A_a = A_{a1} - a_{a1} = A_{a1} - D \tan \alpha$$

B_b = true reading.

B_{b1} = Observed staff reading on B

$$\text{Error} = B_{b1} - B_b = b_{b1} = D \tan \alpha$$

$$\text{True staff reading} = B_b = B_{b1} - b_{b1} = B_{b1} - D \tan \alpha$$

$$\text{True diff in level} = A_a - B_b$$

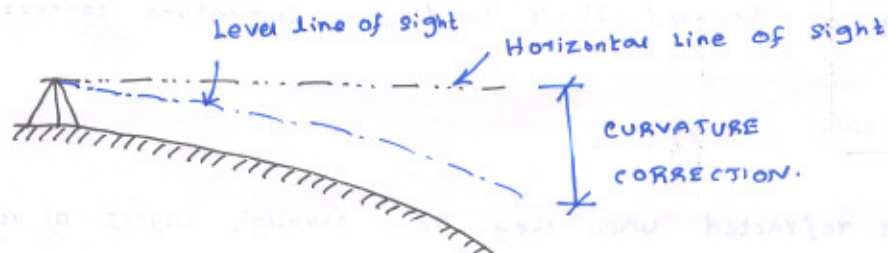
$$= (A_{a1} - D \tan \alpha) - (B_{b1} - D \tan \alpha)$$

$$\boxed{\text{True diff in level} = A_{a1} - B_{b1}}$$

CORRECTIONS TO BE APPLIED:

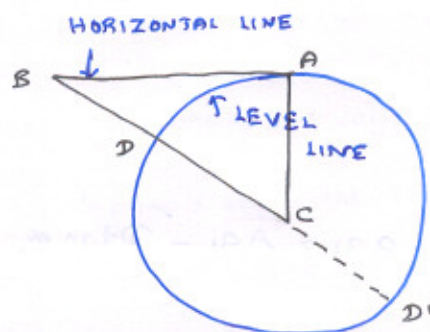
Curvature Correction:

- For long sights, the curvature of the earth affects staff readings.
- The line of sight is horizontal, but the level line is curved and parallel to the mean spheroidal surface of the earth.



The vertical distance between the line of sight and the level line at a particular place is called the curvature correction. Due to curvature, objects appear lower than they really are.

- Curvature correction is always subtractive (ie negative)



$AB = D =$ Horizontal Distance in Kms.

$BD = C_c =$ Curvature correction

$DC = AC = R =$ Radius of Earth

$DD' =$ Diameter [12,742 Km]

Right-Angled Triangle: ABC:

$$BC^2 = AC^2 + AB^2$$

$$(R + C_c)^2 = R^2 + D^2$$

$$R^2 + 2RC_c + C_c^2 = R^2 + D^2$$

$$C_c = \frac{D^2}{2R}$$

C_c^2 is neglected as it is very small in comparison to the diameter of the earth.

$$C_c = \frac{D^2 \times 1000}{12742} = 0.0785 D^2 \text{ m (negative)}$$

True Staff reading = Observed Staff reading - Curvature correction.

REFRACTION CORRECTION:

- Rays of light are refracted when they pass through layers of air of varying density.
- When long sights are taken, line of sight is refracted towards the surface of the earth.
- The refraction curve is irregular because of varying atmospheric conditions.

$$C_r = \frac{1}{7} \frac{d^2}{2R} \text{ (+ve)} = 0.0112 D^2 \text{ m (positive).}$$

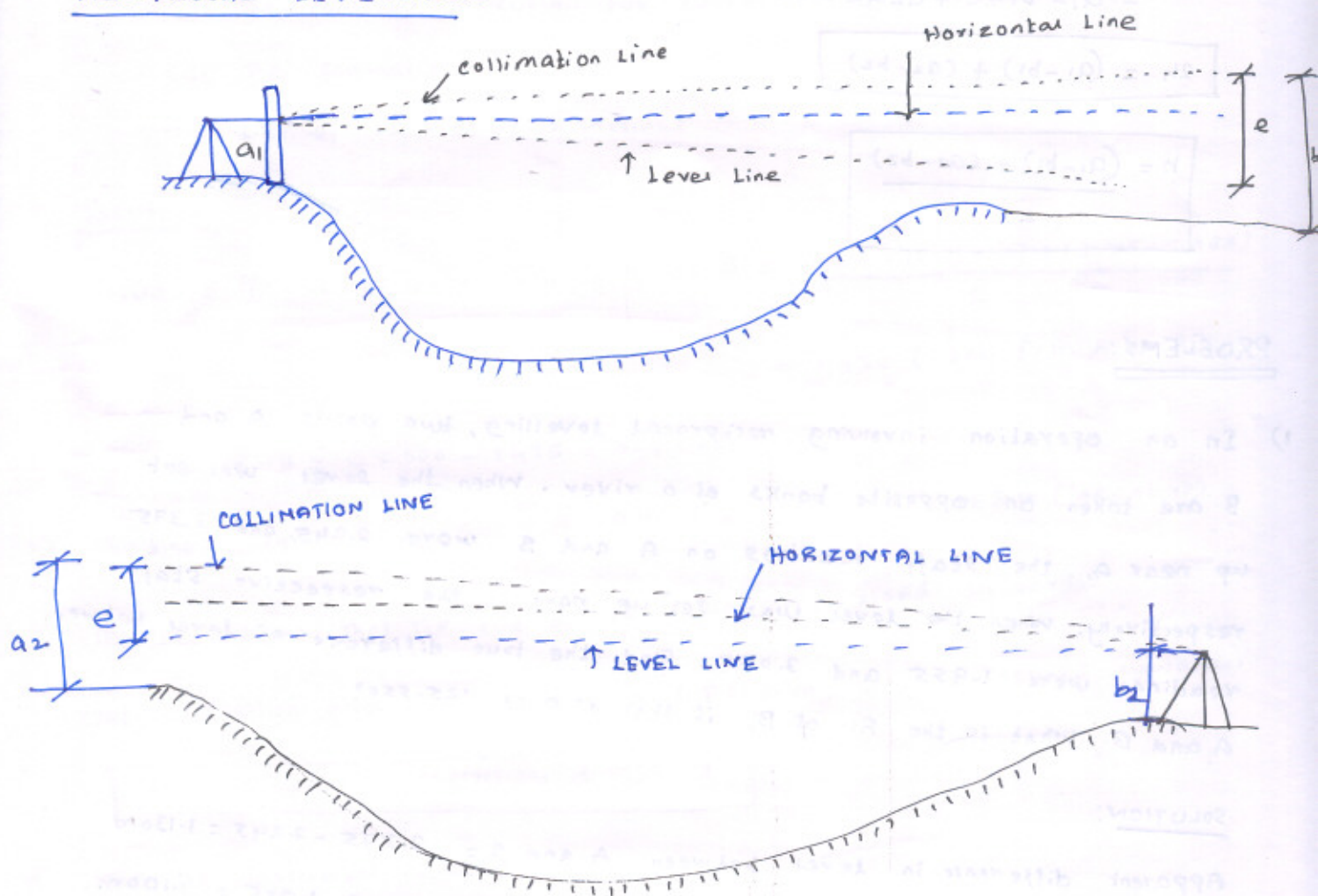
Refraction correction is always additive:

COMBINED CORRECTION:

$$C = \frac{d^2}{2R} - \frac{1}{7} \frac{d^2}{2R} = \frac{6}{7} \frac{d^2}{2R} = -0.0673 D^2 \text{ m}$$

$$C = -0.0673 D^2 \text{ m}$$

RECIPROCAL LEVELLING:



First case:-

Correct staff reading at A = a_1 [as level is very near A]

Correct staff reading at B = $b_1 - e$

True difference of level between A and B

$$h = a_1 - (b_1 - e) \text{ [Fall from B to A]}$$

In the second case:-

Correct staff reading at B = b_2 (as level is near B)

Correct staff reading at A = $a_2 - e$

True difference of level

$$h = (a_2 - e) - b_2$$

$$2h = a_1 - (b_1 - e) + (a_2 - e) - b_2$$

$$= a_1 - b_1 + e + a_2 - e - b_2$$

$$2h = (a_1 - b_1) + (a_2 - b_2)$$

$$h = \frac{(a_1 - b_1) + (a_2 - b_2)}{2}$$

PROBLEMS:

- 1) In an operation involving reciprocal levelling, two points A and B are taken on opposite banks of a river. When the level was set up near A, the staff readings on A and B were 2.245, and 3.375 respectively. When the level was set up near B, the respective staff readings were 1.955 and 3.055. Find the true difference of level between A and B. What is the RL of B, if that of A is 125.550?

SOLUTION:

Apparent difference in level between A and B = $3.375 - 2.245 = 1.130\text{m}$

Apparent difference in level between A and B = $3.055 - 1.955 = 1.100\text{m}$

True difference in level = $\frac{1.130 + 1.100}{2} = 1.115\text{m}$

$$\text{RL of B} = 125.550 - 1.115 = 124.435\text{m}$$